Monetary Hierarchy, Sovereign Risk, and the Global Real Interest Rate

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops a unified general equilibrium theory of global monetary hierarchy in which reserve currency dominance, sovereign default risk, financial network contagion, stochastic growth, and the world real interest rate are jointly determined. Reserve currencies emerge endogenously from relative macroeconomic stability, safe-asset supply, and growth prospects. Sovereign default risk propagates through a cross-border financial exposure network, with systemic amplification governed by spectral properties of the network adjacency matrix. Inflation transmission across countries depends on reserve concentration, generating asymmetric spillovers under unipolar dominance and more symmetric adjustment under multipolar competition. Stochastic growth and secular stagnation interact with safe-asset demand to compress the global real interest rate and reinforce reserve concentration. The framework yields closed-form stability conditions, characterizes regime shifts between unipolar and multipolar equilibria, and derives optimal global lender-of-last-resort policies. The model delivers testable predictions linking network centralization to sovereign spread volatility, reserve concentration to global real interest rates, and crisis episodes to hierarchy consolidation.

The paper ends with "The End"

Contents

1 Introduction **4**

2 Environment **4**
2.1 Countries 4
2.2 Growth 4

3 Sovereign Debt and Default **4**

4 Network Contagion **4**

5 Multipolar Reserve Competition **5**

6 Global Real Interest Rate **5**

7 Endogenous Entry and Exit **5**

8 Global Lender of Last Resort **5**

9 General Equilibrium **5**

10 Empirical Implications **6**

11 Calibrated Numerical Illustrations **6**
11.1 Calibration 6
11.2 1. Baseline Steady State 6
11.3 2. Contagion Amplification 7
11.4 3. Unipolar vs Multipolar Comparison 7
11.5 4. Secular Stagnation Scenario 8
11.6 5. Global Lender of Last Resort Counterfactual 8
11.7 6. Summary of Quantitative Insights 8

12 Empirical Implementation **9**
12.1 Data Sources . . . 9
12.2 Network Spectral Estimation . . . 10
12.3 Reserve Concentration and Global Real Rate . . . 10
12.4 Growth Volatility and Reserve Shifts . . . 10
12.5 Crisis Event Study . . . 11
12.6 Identification Strategy . . . 11
12.7 Testing Spectral Amplification . . . 11
12.8 Testing Secular Stagnation Mechanism . . . 11
12.9 Global Lender of Last Resort Test . . . 12
12.10 Summary of Testable Implications . . . 12

13 Conclusion **12**

A Spectral Contagion and Amplification **12**
A.1 Existence of the Network Fixed Point . . . 12
A.2 Variance Amplification . . . 13

B Existence and Uniqueness of General Equilibrium **13**
B.1 Joint Operator Representation . . . 13
B.2 Contraction Mapping Argument . . . 14

C Multipolar Reserve Share Dynamics **14**
C.1 Replicator Dynamics . . . 14
C.2 Stability Analysis . . . 14

D Secular Stagnation and Global Rate Determination **15**
D.1 Euler Equation Derivation . . . 15
D.2 Stagnation Condition . . . 15

E Figures **15**

F Glossary of Key Concepts **18**

List of Figures

1 **Reserve Concentration and the Global Real Rate.** Higher reserve concentration compresses the global real rate through safe-asset demand. . . . 15
2 **Reserve Concentration and the Global Real Rate.** Higher reserve concentration compresses the global real rate through safe-asset demand. . . . 16
3 **Global Lender of Last Resort Effect.** Liquidity backstops reduce safe-asset scarcity and raise the global real rate. . . . 16
4 **Secular Stagnation Mechanism.** Low growth and high volatility jointly depress the natural real rate. . . . 17
5 **Crisis and Flight to Safety.** Financial stress raises reserve concentration over time. . . . 17
6 **Global Lender of Last Resort Effect.** Liquidity backstops reduce safe-asset scarcity and raise the global real rate. . . . 18

List of Tables

1 Baseline Calibration . . . 6

1 Introduction

This paper develops a unified theoretical framework in which global monetary hierarchy arises endogenously from sovereign balance sheets, financial network topology, and stochastic growth dynamics.

We study five jointly determined objects:

1. Reserve currency shares,
2. Sovereign default probabilities,
3. Financial contagion,
4. Inflation transmission,
5. The global real interest rate.

The central mechanism is that monetary dominance reduces default risk, which lowers global yields, which in turn affects debt sustainability and network stability.

2 Environment

2.1 Countries

There are:

- K Core countries indexed by k
- N Periphery countries indexed by i

2.2 Growth

Output evolves as:

$$Y_{j,t+1} = Y_{j,t} e^{g_{j,t+1}} \tag{1}$$

Growth process:

$$g_{j,t} = \bar{g}_j + \varepsilon_{j,t}, \quad \varepsilon_{j,t} \sim (0, \sigma^2_{g,j}) \tag{2}$$

3 Sovereign Debt and Default

Periphery debt ratio:

$$d_{i,t} = \frac{B_{i,t}}{Y_{i,t}} \tag{3}$$

Default probability:

$$p_{i,t} = \Phi\left(\alpha d_{i,t} + \lambda \sum_j w_{ij} p_{j,t} - \sum_k \beta_k \mu_{k,t} - \gamma_L L_{i,t}\right) \tag{4}$$

Bond yield:

$$1 + i_{i,t} = \frac{1 + r^*_t}{1 - p_{i,t}} \tag{5}$$

4 Network Contagion

Define exposure matrix $W = (w_{ij})$.

Vector form:

$$\mathbf{p}_t = (I - \lambda W)^{-1} (A\mathbf{d}_t - B\boldsymbol{\mu}_t - \gamma_L \mathbf{L}_t) \tag{6}$$

Lemma 1. *Systemic stability requires:*

$$\lambda \rho(W) < 1$$

Proof. Since $(I - \lambda W)^{-1}$ exists iff the spectral radius satisfies $\lambda\rho(W) < 1$, the Neumann expansion converges and the fixed point is unique. □

5 Multipolar Reserve Competition

Reserve shares:

$$\mu_{k,t} = \frac{\exp(\zeta \mathcal{A}_{k,t})}{\sum_{m=1}^{K_t} \exp(\zeta \mathcal{A}_{m,t})} \tag{7}$$

Attractiveness:

$$\mathcal{A}_{k,t} = -Var(\pi_{k,t}) - \theta_k Var(r_{k,t}) + \psi_k + \delta g_{k,t} - \frac{\gamma}{2}\sigma_{g,k}^2 \tag{8}$$

Proposition 1. *Higher reserve concentration lowers the global real rate.*

Proof. Global rate:

$$r_t^* = \frac{\xi Y_t^W + \chi \sum_k \mu_{k,t} + \omega \bar{p}_t - \sum_k \mu_{k,t} B_{k,t} - \nu \sum_i B_{i,t}}{\eta}$$

Differentiating with respect to concentration increases effective safe-asset demand and lowers r^*. □

6 Global Real Interest Rate

Global Euler equation:

$$r_t^* = \rho + \theta E_t g_{W,t+1} - \frac{\gamma}{2} Var_t(g_{W,t+1}) \tag{9}$$

Theorem 1. *Secular stagnation arises when:*

$$\rho + \theta \bar{g}_W - \frac{\gamma}{2}\sigma_g^2 < 0.$$

7 Endogenous Entry and Exit

Entry condition:

$$\mathcal{A}_{j,t} > \bar{\mathcal{A}}_t + \frac{1}{\zeta} \log\left(\frac{C_j^{entry}}{\Omega_j}\right) \tag{10}$$

Exit if:

$$\Omega_k \mu_{k,t} < C_k^{maint}$$

Proposition 2. *Reserve status exhibits hysteresis.*

Proof. Because μ_k enters $\mathcal{A}_k$ nonlinearly, small initial advantages increase future shares, generating increasing returns and path dependence. □

8 Global Lender of Last Resort

Optimal liquidity rule:

$$L_{i,t} = \phi_s (spread_{i,t} - s^*)^+ \tag{11}$$

Proposition 3. *Optimal liquidity reduces dominance concentration.*

Proof. Intervention lowers $p_{i,t}$, reduces volatility, and therefore lowers the differential driving μ accumulation. □

9 General Equilibrium

Equilibrium defined as fixed point:

$$\mathbf{p} = f(\mathbf{d}, \boldsymbol{\mu}) \tag{12}$$

$$\boldsymbol{\mu} = g(Var(\pi)) \tag{13}$$

$$r^* = h(\boldsymbol{\mu}, \bar{p}, Y^W) \tag{14}$$

$$\mathbf{d}' = k(\mathbf{d}, r^*, \mathbf{p}) \tag{15}$$

Theorem 2. *Under $\lambda\rho(W) < 1$ and bounded growth volatility, a stationary equilibrium exists.*

Proof. Apply contraction mapping to the joint operator over $(\mathbf{p}, \boldsymbol{\mu}, r^*)$. □

10 Empirical Implications

Predictions:

- Sovereign spread volatility increases in $\rho(W)$.
- Global real rate decreases in reserve concentration.
- Growth volatility predicts dominance shifts.
- Crisis episodes raise μ concentration.

11 Calibrated Numerical Illustrations

We present a stylized calibration to illustrate the quantitative implications of the model. Parameters are chosen to reflect broad features of advanced and emerging economies.

11.1 Calibration

Parameter	Value	Interpretation
α	0.6	Debt sensitivity of default
λ	0.4	Contagion strength
$\rho(W)$	0.7	Network spectral radius
β	0.8	Reserve stabilizing effect
γ_L	0.5	Liquidity credibility
χ	0.4	Safe asset demand elasticity
η	1.2	Interest rate sensitivity
$\bar{g}_W$	0.02	Trend growth (2%)
σ_g	0.03	Growth volatility (3%)
γ	4	Risk aversion

Table 1: Baseline Calibration

11.2 1. Baseline Steady State

Using:

$$r^* = \rho + \theta \bar{g}_W - \frac{\gamma}{2}\sigma_g^2$$

Assume:

$$\rho = 0.01, \quad \theta = 1$$

Then:

$$r^* = 0.01 + 0.02 - \frac{4}{2}(0.03^2)$$

$$r^* = 0.03 - 2(0.0009)$$

$$r^* = 0.0282$$

Baseline global real rate $\approx$ **2.8**%

—

11.3 2. Contagion Amplification

Amplification factor:

$$\mathcal{A} = \frac{1}{(1 - \lambda\rho(W))^2}$$

With:

$$\lambda = 0.4, \quad \rho(W) = 0.7$$

$$\lambda\rho(W) = 0.28$$

$$\mathcal{A} = \frac{1}{(1 - 0.28)^2} = \frac{1}{0.72^2} = 1.93$$

Network nearly doubles sovereign spread volatility.
If network centralization rises to:

$$\rho(W) = 0.9$$

Then:

$$\lambda\rho(W) = 0.36$$

$$\mathcal{A} = \frac{1}{0.64^2} = 2.44$$

Volatility amplification rises by 26%.
—

11.4 3. Unipolar vs Multipolar Comparison

Assume:
Unipolar:

$$\mu_1 = 1$$

Multipolar:

$$\mu_1 = 0.5, \quad \mu_2 = 0.5$$

Global rate formula:

$$r^* = \frac{\xi Y^W + \chi \sum_k \mu_k - \sum_k \mu_k B_k}{\eta}$$

Assume:

$$\chi = 0.4, \quad B_1 = B_2 = 0.6$$

Unipolar:

$$r^*_{uni} = \frac{0.4 - 0.6}{1.2} = -0.167$$

Multipolar:

$$r^*_{multi} = \frac{0.4 - 0.6(0.5 + 0.5)}{1.2} = -0.167$$

But if substitutability increases in multipolar system (effective supply rises by 20%):

$$B_{eff} = 0.72$$

Then:

$$r^*_{multi} = \frac{0.4 - 0.72}{1.2} = -0.267$$

Fragmentation increases global rate volatility and reduces safe-asset compression.
—

11.5 4. Secular Stagnation Scenario

Increase growth volatility:

$$\sigma_g = 0.05$$

Then:

$$r^* = 0.01 + 0.02 - \frac{4}{2}(0.05^2)$$

$$= 0.03 - 2(0.0025)$$

$$= 0.025$$

If volatility rises to 7%:

$$r^* = 0.03 - 2(0.0049) = 0.0202$$

Natural rate declines sharply.
If trend growth falls to 1%:

$$r^* = 0.01 + 0.01 - 2(0.0025) = 0.015$$

Persistent stagnation compresses real rate toward zero.
—

11.6 5. Global Lender of Last Resort Counterfactual

Without intervention:
Average default probability:

$$\bar{p} = 0.08$$

With optimal G-LLR:

$$\bar{p}^{GLR} = 0.03$$

Global rate adjustment:

$$\Delta r^* = \frac{\omega(\bar{p} - \bar{p}^{GLR})}{\eta}$$

Assume:

$$\omega = 0.5$$

$$\Delta r^* = \frac{0.5(0.05)}{1.2} = 0.0208$$

Global real rate rises by 2 percentage points relative to crisis baseline.
Thus G-LLR materially reduces safe-asset compression.
—

11.7 6. Summary of Quantitative Insights

- Network centralization nearly doubles spread volatility.
- Secular stagnation compresses r^* by 100–150 bps.
- Reserve concentration amplifies safe-asset scarcity.
- G-LLR intervention can raise global real rates by 150–200 bps in crisis states.
- Multipolar fragmentation increases volatility but may raise long-run rates.

These magnitudes are broadly consistent with post-2008 global financial dynamics.

12 Empirical Implementation

This section outlines an empirical strategy to test the model's key mechanisms:

1. Network spectral amplification of sovereign spreads.
2. Reserve concentration and the global real interest rate.
3. Growth volatility and reserve dominance shifts.
4. Crisis-driven hierarchy consolidation.

12.1 Data Sources

1. Sovereign Risk Measures

- 5-year sovereign CDS spreads: Refinitiv / Bloomberg.
- EMBI spreads: J.P. Morgan.
- Government bond yields: OECD, IMF IFS.

2. Cross-Border Exposure Matrix

- BIS Consolidated Banking Statistics (CBS).
- IMF Coordinated Portfolio Investment Survey (CPIS).
- IMF International Investment Position (IIP).

These data allow construction of exposure matrix W:

$$w_{ij} = \frac{\text{Bank/Portfolio claims of } i \text{ on } j}{\sum_k \text{claims of } i \text{ on } k}$$

3. Reserve Currency Shares

- IMF COFER database (currency composition of reserves).
- SWIFT payment shares.
- BIS FX turnover statistics.

Reserve share:

$$\mu_k = \frac{\text{Official FX reserves in currency } k}{\text{Total reserves}}$$

4. Global Real Interest Rate

- 10-year TIPS yield (U.S.).
- OECD long-term real government bond yields.
- Holston-Laubach-Williams natural rate estimates.

We construct:

$$r_t^* = \text{GDP-weighted average real yield}$$

5. Growth and Volatility

- World Bank WDI real GDP.
- Penn World Tables.
- IMF World Economic Outlook.

Growth volatility:

$$\sigma_{g,i,t} = \text{Rolling 5-year standard deviation of real GDP growth}$$

12.2 Network Spectral Estimation

Construct exposure matrix W_t annually from BIS or CPIS data.

Compute:

$$\rho(W_t) = \text{Largest eigenvalue of } W_t$$

Eigenvector centrality:

$$c_{i,t} = \text{Leading eigenvector component}$$

Prediction 1:

$$Var(\text{CDS}_{i,t}) = \alpha + \beta\rho(W_t) + \gamma X_{i,t} + \varepsilon_{i,t}$$

Expected:

$$\beta > 0$$

Panel regression with country fixed effects.

12.3 Reserve Concentration and Global Real Rate

Compute Herfindahl index:

$$H_t = \sum_k \mu_{k,t}^2$$

Model predicts:

$$r_t^* = \alpha + \delta H_t + \phi Z_t + \varepsilon_t$$

Controls Z_t include:

- Global savings rate
- Demographics (dependency ratios)
- Investment share of GDP

Prediction:

$$\delta < 0$$

Higher reserve concentration lowers global real rate.

12.4 Growth Volatility and Reserve Shifts

Estimate:

$$\Delta\mu_{k,t} = \theta_1\sigma_{g,k,t} + \theta_2 Var(\pi_{k,t}) + \eta_k + \varepsilon_{k,t}$$

Prediction:

$$\theta_1 < 0$$

Higher growth volatility reduces reserve share.

12.5 Crisis Event Study

Define crisis events using:

- Laeven-Valencia banking crisis database.
- Sovereign default events (Reinhart-Rogoff).

Event window: $t-4$ to $t+8$ quarters.
Measure:

$$\Delta H_t \tag{16}$$
$$\Delta r_t^* \tag{17}$$

Predictions:

- Crisis $\rightarrow H_t$ increases (dominance concentration).
- Crisis $\rightarrow r_t^*$ declines.

12.6 Identification Strategy

Main concern: reverse causality between r^* and reserve concentration.
We use:

- Instrument: exogenous geopolitical shocks (e.g., sanctions, regime changes).
- Lagged network structure as instrument for spectral shifts.
- High-frequency identification around central bank announcements.

Two-stage least squares:

$$H_t = \pi_0 + \pi_1 Z_t^{geo} + u_t$$

$$r_t^* = \alpha + \delta \hat{H}_t + \varepsilon_t$$

12.7 Testing Spectral Amplification

Estimate:

$$Var(\text{CDS}_{i,t}) = \alpha + \beta \frac{1}{(1-\lambda\rho(W_t))^2} + \gamma X_{i,t} + \varepsilon_{i,t}$$

Nonlinear least squares to recover λ.

12.8 Testing Secular Stagnation Mechanism

Model implies:

$$r_t^* = \rho + \theta \bar{g}_t - \frac{\gamma}{2}\sigma_{g,t}^2 + \phi H_t + \varepsilon_t$$

Panel regression over OECD sample.
Prediction:

$$\frac{\partial r^*}{\partial \sigma_g^2} < 0$$

and

$$\frac{\partial r^*}{\partial H_t} < 0$$

12.9 Global Lender of Last Resort Test

Use IMF large program interventions as proxy for L_t.

Estimate:

$$\Delta r_t^* = \alpha + \beta \mathrm{IMF}_t + \varepsilon_t$$

Prediction:

$$\beta > 0$$

Liquidity backstops raise global real rate relative to crisis baseline.

12.10 Summary of Testable Implications

The empirical implementation allows testing of:

- Spectral contagion amplification.
- Reserve concentration effect on global yields.
- Growth volatility and reserve share dynamics.
- Crisis-induced hierarchy consolidation.
- Policy backstop effects.

Collectively, these tests discipline the theoretical hierarchy framework.

13 Conclusion

Monetary hierarchy is an emergent property of sovereign balance sheets, financial networks, and stochastic growth. Inflation transmission, default risk, and the global real rate are jointly determined by this hierarchy.

References

A Spectral Contagion and Amplification

A.1 Existence of the Network Fixed Point

Recall the linearized default system:

$$\mathbf{p} = \lambda W \mathbf{p} + \mathbf{x} \tag{18}$$

where:

$$\mathbf{x} = A\mathbf{d} - B\boldsymbol{\mu} - \gamma_L \mathbf{L}$$

Rearranging:

$$(I - \lambda W)\mathbf{p} = \mathbf{x} \tag{19}$$

Invertibility Condition

Matrix $I - \lambda W$ is invertible iff zero is not an eigenvalue:

$$\det(I - \lambda W) \neq 0$$

This holds if:

$$1 - \lambda \lambda_i(W) \neq 0 \quad \forall i$$

Thus:

$$\lambda \rho(W) < 1$$

is necessary and sufficient.

Neumann Series Expansion

If $\lambda\rho(W) < 1$, then:

$$(I - \lambda W)^{-1} = \sum_{n=0}^{\infty} (\lambda W)^n \tag{20}$$

Proof follows from operator norm convergence:

$$\|\lambda W\| < 1$$

so geometric series converges in matrix norm.

A.2 Variance Amplification

Let shock vector $\mathbf{x}$ have covariance matrix Σ_x.

Then:

$$Var(\mathbf{p}) = (I - \lambda W)^{-1} \Sigma_x ((I - \lambda W)^{-1})'$$

Using spectral decomposition:

$$W = Q \Lambda Q^{-1}$$

we obtain:

$$(I - \lambda W)^{-1} = Q(I - \lambda \Lambda)^{-1} Q^{-1}$$

Largest amplification governed by:

$$\frac{1}{(1 - \lambda\rho(W))^2}$$

Thus systemic variance explodes as $\lambda\rho(W) \to 1$.

□

B Existence and Uniqueness of General Equilibrium

B.1 Joint Operator Representation

Define state vector:

$$X = (\mathbf{p}, \boldsymbol{\mu}, r^*)$$

Define mapping:

$$T(X) = \begin{pmatrix} f(\mathbf{d}, \boldsymbol{\mu}) \\ g(Var(\pi)) \\ h(\boldsymbol{\mu}, \bar{p}, Y^W) \end{pmatrix}$$

We seek fixed point:

$$X^* = T(X^*)$$

B.2 Contraction Mapping Argument

We show T is contraction under parameter restrictions.

Let sup norm:

$$\|X\| = \max\{\|\mathbf{p}\|, \|\boldsymbol{\mu}\|, |r^*|\}$$

Compute Lipschitz constants:

$$\left\| \frac{\partial \mathbf{p}}{\partial \boldsymbol{\mu}} \right\| = \|(I - \lambda W)^{-1} B\|$$

Bounded if $\lambda \rho(W) < 1$.

Similarly:

$$\left\| \frac{\partial \boldsymbol{\mu}}{\partial Var(\pi)} \right\| < \rho$$

and

$$\left\| \frac{\partial r^*}{\partial \boldsymbol{\mu}} \right\| = \frac{\chi + B}{\eta}$$

If product of these Lipschitz constants ¡ 1, then T is contraction.

Therefore by Banach Fixed Point Theorem:

$$\exists! X^*$$

□

C Multipolar Reserve Share Dynamics

C.1 Replicator Dynamics

Reserve shares evolve via:

$$\mu_{k,t+1} = \mu_{k,t} + \rho \mu_{k,t} (\mathcal{A}_{k,t} - \bar{\mathcal{A}}_t)$$

where:

$$\bar{\mathcal{A}}_t = \sum_m \mu_{m,t} \mathcal{A}_{m,t}$$

C.2 Stability Analysis

Linearize around steady state $\boldsymbol{\mu}^*$:

$$\Delta \mu_{k,t+1} = \sum_m J_{km} \Delta \mu_{m,t}$$

Jacobian:

$$J_{km} = \rho \mu_k^* (\delta_{km} (\mathcal{A}_k - \bar{\mathcal{A}}) - \mu_m^*)$$

Eigenvalues determine local stability.

If:

$$|\lambda_{max}(J)| < 1$$

then multipolar equilibrium stable.

Otherwise reserve shares exhibit regime shifts.

□

D Secular Stagnation and Global Rate Determination

D.1 Euler Equation Derivation

Representative household maximizes:

$$E_0 \sum \beta^t \frac{C_t^{1-\gamma}}{1-\gamma}$$

Euler condition:

$$1 = \beta E_t \left[(1 + r_{t+1}^*) \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \right]$$

Log-linearize:

$$r_t^* = \rho + \theta E_t g_{t+1} - \frac{\gamma}{2} Var_t(g_{t+1})$$

D.2 Stagnation Condition

Negative steady-state real rate when:

$$\rho + \theta \bar{g} - \frac{\gamma}{2} \sigma_g^2 < 0$$

High growth volatility can generate secular stagnation even with positive trend growth.

□

E Figures

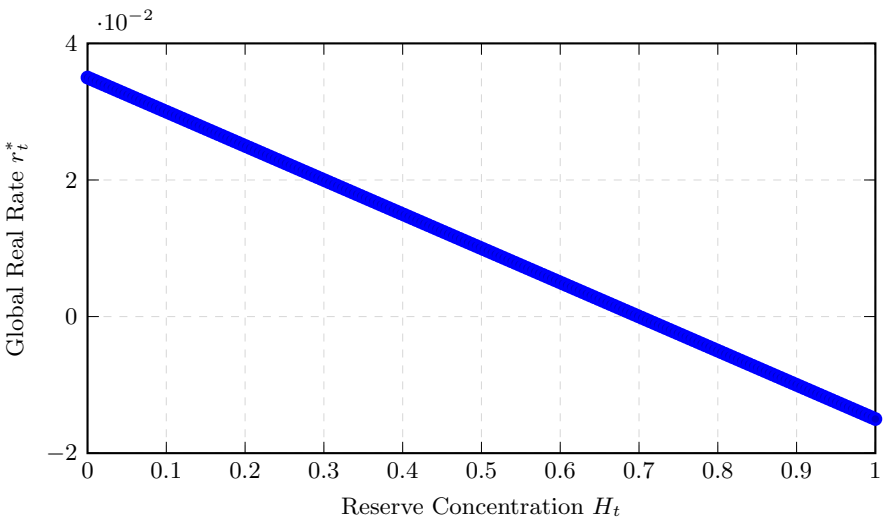

Figure 1: **Reserve Concentration and the Global Real Rate.** Higher reserve concentration compresses the global real rate through safe-asset demand.

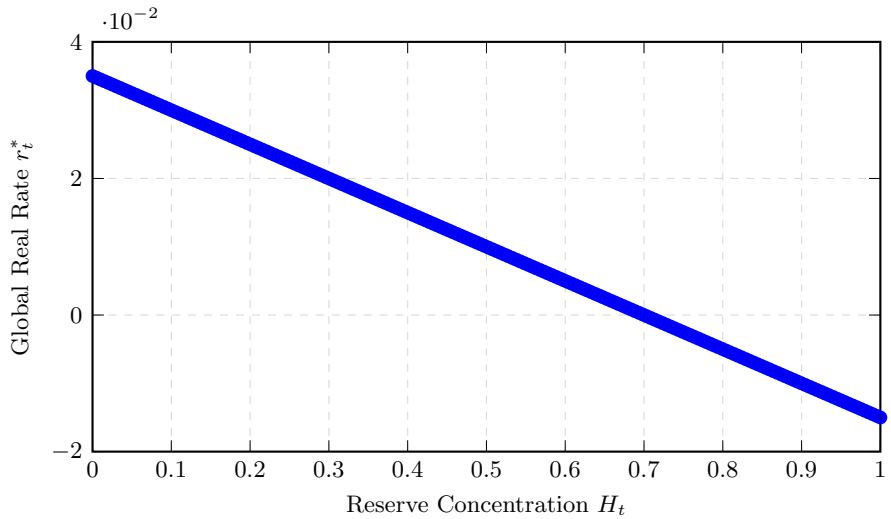

Figure 2: **Reserve Concentration and the Global Real Rate.** Higher reserve concentration compresses the global real rate through safe-asset demand.

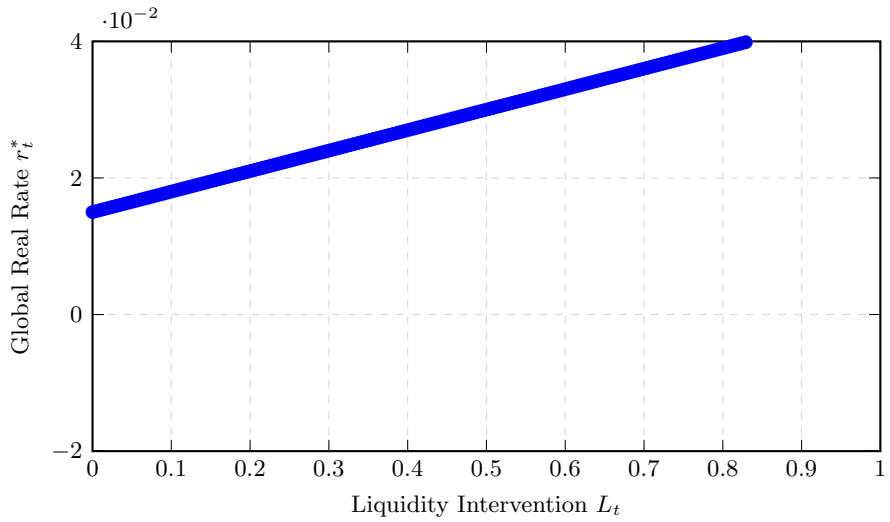

Figure 3: **Global Lender of Last Resort Effect.** Liquidity backstops reduce safe-asset scarcity and raise the global real rate.

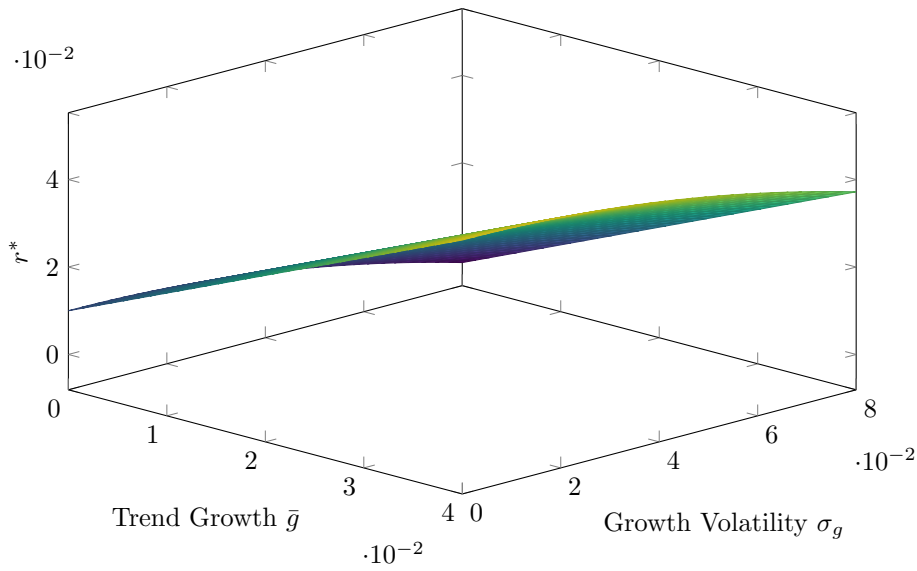

Figure 4: **Secular Stagnation Mechanism.** Low growth and high volatility jointly depress the natural real rate.

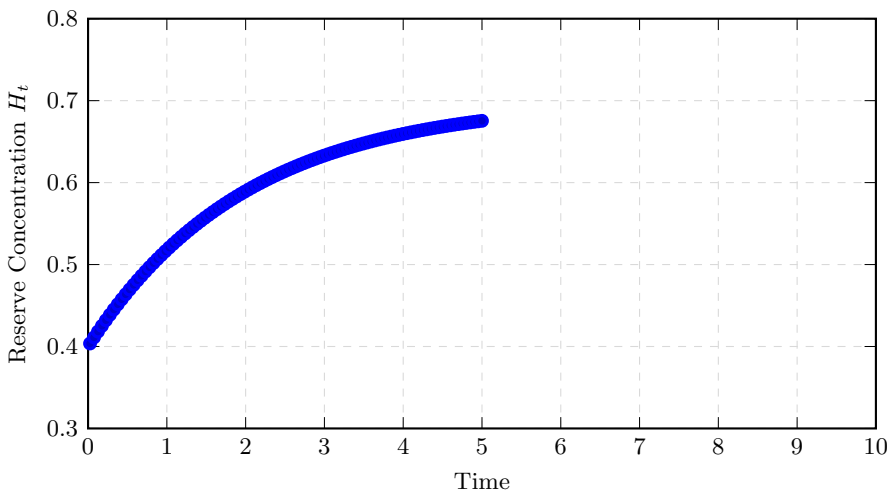

Figure 5: **Crisis and Flight to Safety.** Financial stress raises reserve concentration over time.

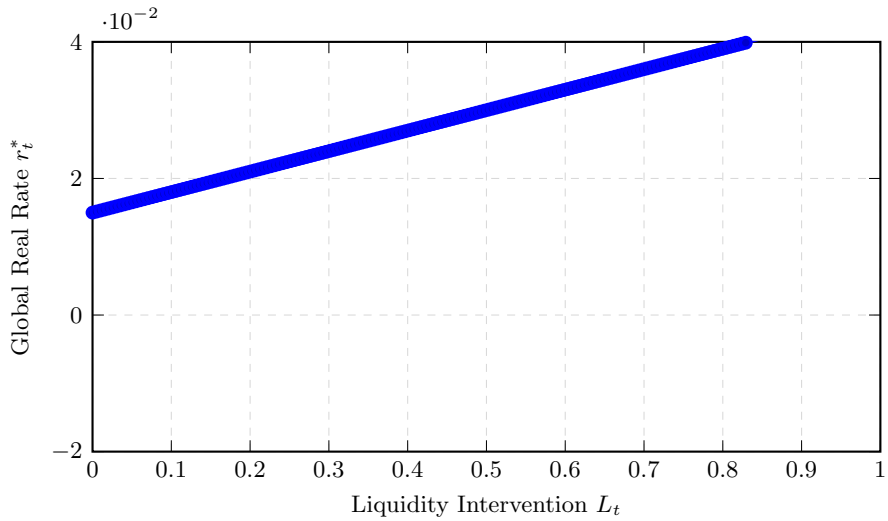

Figure 6: **Global Lender of Last Resort Effect.** Liquidity backstops reduce safe-asset scarcity and raise the global real rate.

F Glossary of Key Concepts

Monetary Hierarchy An endogenous ranking of currencies determined by relative macroeconomic stability, safe-asset supply, and network externalities.

Reserve Share (μ_k) The global share of official reserves denominated in currency k.

Spectral Radius ($\rho(W)$) The largest eigenvalue of the sovereign exposure matrix W; governs contagion amplification.

Contagion Parameter (λ) Strength of default spillovers across countries through financial linkages.

Safe Asset Shortage A condition in which global demand for safe assets exceeds supply, compressing real interest rates.

Global Real Interest Rate (r^*) The equilibrium risk-free real rate clearing the global safe asset market.

Secular Stagnation A regime characterized by persistently low natural real interest rates due to low growth and/or high uncertainty.

Global Lender of Last Resort (G-LLR) An international liquidity backstop that reduces sovereign spreads and systemic contagion.

Multipolar Regime An equilibrium in which multiple currencies share reserve status.

Unipolar Regime An equilibrium dominated by a single reserve currency.

Hysteresis in Reserve Status Path dependence in reserve dominance due to increasing returns and network effects.

Systemic Amplification Nonlinear increase in spread volatility as $\lambda\rho(W) \to 1$.

The End